Equality in the Product-Height Inequality for Elliptic Abscissas

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Abstract

For a fixed elliptic curve over $\mathbb{Q}(\sqrt{2})$, the logarithmic height of a product of abscissas of nonzero multiples of a specified point equals the sum of their heights if and only if the indices have the same 2-adic valuation. Positive integer weights are allowed. We prove both directions by reduction at finite primes and the product formula, without a rank calculation or a conjectural arithmetic input. This extends the odd-index identity used in Sun's twenty-eight-unknown construction. Mixed valuations have a positive height defect; the pair of indices $1, 2$ has defect exactly $\log 41$. We also give a number-field criterion and a three-scale height consequence.

1 Statement and height convention

Put $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$, $d = 3 - \rho$, $D = d^2$, and

$$E : y^2 = x^3 - Dx, \qquad R_0 = (1 + \rho, 2\rho), \qquad P = 2R_0 = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32}\right).$$

Write h for the absolute logarithmic Weil height. For a number field F containing $a \neq 0$, our convention is

$$[F : \mathbb{Q}]h(a) = \sum_{\sigma : F \hookrightarrow \mathbb{C}} \max(0, \log|\sigma(a)|) + \sum_{\mathfrak{p}} \max(0, -\operatorname{ord}_{\mathfrak{p}} a) \log \mathrm{N}\,\mathfrak{p}.$$

Both complex embeddings of a conjugate pair occur in the first sum. The order at a prime ideal has value group $\mathbb{Z}$.

Theorem 1.1. *The point P has infinite order. For nonzero integers $n_1, \dots, n_s$ and positive integers $a_1, \dots, a_s$, put $x_i = x([n_i]P)$. Then*

$$h\left(\prod_{i=1}^{s} x_i^{a_i}\right) = \sum_{i=1}^{s} a_i h(x_i) \quad \Longleftrightarrow \quad v_2(n_1) = \cdots = v_2(n_s).$$

If the valuations differ, the difference between the right and left sides is at least $\log 3$.

The odd-index special case is the identity stated in [1, Lemma 2.2], where it is attributed to the earlier construction. Here equality is characterized, rather than merely established for odd indices. Signs and repetitions of the indices are unrestricted. No assertion that $2E(K) = \mathbb{Z}P$ is needed.

2 An exact expression for the height defect

For an embedding σ, set $L_\sigma(a) = \log|\sigma(a)|$. For a finite prime set $L_{\mathfrak{p}}(a) = -\operatorname{ord}_{\mathfrak{p}}(a) \log \mathrm{N}\,\mathfrak{p}$. Thus $[F : \mathbb{Q}]h(a) = \sum_v \max(0, L_v(a))$.

Lemma 2.1. *For nonzero $u_i \in F$ and positive real weights a_i, the formal weighted height defect is*

$$\sum_i a_i h(u_i) - \frac{1}{[F:\mathbb{Q}]} \sum_v \max\left(0, \sum_i a_i L_v(u_i)\right) = \frac{1}{[F:\mathbb{Q}]} \sum_v \min(P_v, N_v),$$

where $P_v = \sum_i a_i \max(0, L_v(u_i))$ and $N_v = \sum_i a_i \max(0, -L_v(u_i))$. For integral weights the second term is the height of the product. Equality holds exactly when no place has both a strictly positive and a strictly negative value among the $L_v(u_i)$.

Proof. At each place $\sum_i a_i L_v(u_i) = P_v - N_v$, and $P_v - \max(0, P_v - N_v) = \min(P_v, N_v)$. All terms are nonnegative; only finitely many finite primes contribute. □

3 The exceptional primes and the real embeddings

The discriminant of the displayed integral equation is $64D^3$. Its only possibly bad prime ideals are (ρ) and (d).

Lemma 3.1. *The point P is nontorsion. For every $n \neq 0$ one has*

$$\operatorname{ord}_{(\rho)} x([n]P) = -6 - 4v_2(n), \qquad \operatorname{ord}_{(d)} x([n]P) \leq 0.$$

At both real embeddings σ of K, $\sigma(x([n]P)) > \sigma(d) > 1$. Every $x([n]P)$ is a nonzero square in K.

Proof. At (ρ), normalize $\operatorname{ord}(\rho) = 1$, so $\operatorname{ord}(2) = 2$. The explicit coordinates give $\operatorname{ord} x(P) = -6$, $\operatorname{ord} y(P) = -9$, and $\operatorname{ord} z(P) = 3$ for $z = -x/y$. The formal group at the integral point at infinity has an integral invariant differential; its logarithm therefore has coefficients b_{m-1}/m, with b_{m-1} integral. If $\operatorname{ord} z \geq 3$, then for $m \geq 2$

$$\operatorname{ord}(b_{m-1} z^m/m) \geq m \operatorname{ord} z - 2v_2(m) > \operatorname{ord} z,$$

since $3(m-1) > 2v_2(m)$. Thus the logarithm is valuation-preserving on this neighborhood. Integer multiplication preserves the neighborhood, and $\log_E([n]P) = n \log_E(P) \neq 0$ for $n \neq 0$. This proves nontorsion and gives $\operatorname{ord} z([n]P) = 3 + 2v_2(n)$. The integral expansion $x = z^{-2} + O(z^{-1})$ yields the first formula. These formal-group facts apply to the displayed integral equation; minimality at 2 is not required.

At (d) the residue field is $\mathbb{F}_7$, with $\rho = 3$. The equation reduces to $y^2 = x^3$, and P reduces to $(1, 6)$, a nonsingular point. The equation is minimal here: its discriminant has valuation $6 < 12$. Points with nonsingular reduction form a subgroup, so every multiple of P has nonsingular reduction. Positive valuation of its abscissa would force positive valuation of the ordinate and reduction $(0, 0)$, which is singular. This is impossible.

For any nonzero n, the point $Q = [n]R_0$ is nontorsion and has nonzero ordinate. Duplication gives

$$x([n]P) = x(2Q) = \left(\frac{x(Q)^2 + D}{2y(Q)}\right)^2.$$

At both real embeddings the numerator is positive, so this square is strictly positive. The equation $y^2 = x(x^2 - D)$, together with nontorsion, now gives $x > \sqrt{D} = d$ at that embedding. The formal group and nonsingular reduction statements used here are standard for integral Weierstrass equations; see [2, Chapters IV and VII]. □

4 Good reduction and the equality criterion

Lemma 4.1. *At a prime of good reduction for the displayed equation,*

$$\operatorname{ord}_{\mathfrak{p}} x([n]P) > 0 \iff [n]\overline{P} = (0,0), \qquad \operatorname{ord}_{\mathfrak{p}} x([n]P) < 0 \iff [n]\overline{P} = O.$$

If indices n, m give respectively a positive and a negative valuation, then $v_2(n) < v_2(m)$.

Proof. The good prime has odd residue characteristic and D is a unit. An integral point with abscissa zero modulo the prime has ordinate zero; the point $(0,0)$ is nonzero two-torsion. Negative abscissa valuation characterizes reduction to infinity. If t is the finite order of $\overline{P}$, then $[n]\overline{P}$ has order two, whereas $t \mid m$. Thus $v_2(n) = v_2(t) - 1$ and $v_2(m) \geq v_2(t)$. □

Proof of Theorem 1.1. Suppose first that all indices have the same 2-adic valuation. Lemma 4.1 excludes opposite finite valuation signs at a good prime. Lemma 3.1 excludes opposite signs at the two remaining primes and at the real embeddings. Lemma 2.1 proves equality.

Conversely, let $r = \min_i v_2(n_i)$ and suppose some index has larger valuation. Put $Q = [2^r]P$. Both real values of $x(Q)$ exceed one. The product formula forces $\operatorname{ord}_{\mathfrak{p}} x(Q) > 0$ at some finite prime: otherwise all finite logarithms and all archimedean logarithms would be nonnegative, with a positive sum. Lemma 3.1 makes this a good prime. There $[2^r]\overline{P} = (0,0)$, so $\overline{P}$ has order exactly 2^{r+1}. All indices of valuation r therefore reduce to $(0,0)$, and all indices of larger valuation reduce to O. Both signs occur, proving strict inequality.

All abscissas are squares, so the nonzero valuations just found have absolute value at least two. The positive integral weights make $\min(P_{\mathfrak{p}}, N_{\mathfrak{p}}) \geq 2 \log \mathrm{N}\,\mathfrak{p}$. Dividing by $[K : \mathbb{Q}] = 2$ gives a defect at least $\log \mathrm{N}\,\mathfrak{p}$. The prime does not lie over 2, so $\mathrm{N}\,\mathfrak{p} \geq 3$. □

Corollary 4.2 (A number-field criterion)**.** *Let F be a number field, $D \in \mathcal{O}_F \setminus \{0\}$, and $E_D : y^2 = x^3 - Dx$. Let $P \in E_D(F)$ be nontorsion. Assume that for every $n \neq 0$, every embedding σ satisfies $|\sigma(x([n]P))| > 1$, and every prime over $2D$ satisfies $\operatorname{ord}_{\mathfrak{p}} x([n]P) \leq 0$. Then the product-height equality for positive integer weights holds exactly when all the indices have the same 2-adic valuation.*

Proof. Outside $2D$ the displayed equation has good reduction. The two good reduction tests in Lemma 4.1, the sign criterion of Lemma 2.1, and the product-formula argument in the converse apply without change. The quantitative bound $\log 3$ is not asserted in this generality. □

5 A three-scale consequence

Corollary 5.1. *Let nonzero integers $n_1, \ldots, n_s$ have a common 2-adic valuation, and put $C = \prod_i x([n_i]P) = C_0 + C_1\rho$. Set $A_0 = 1$, $A_1 = C_0^2$, $A_2 = (1 + C_1^2)^2$. These are nonzero rational squares. There are constants $c_1, c_2 > 0$ depending only on E, P such that, for $\eta \in \mathbb{Q}^\times$ and $H \geq 1$,*

$$h(A_j\eta) \leq H \quad (0 \leq j \leq 2) \quad \Longrightarrow \quad \max_i |n_i|^2 \leq c_1 H + c_2.$$

The constants are independent of the common valuation, the number of indices, and the indices themselves.

Proof. Both real embeddings of C are positive, so $C_0 > 0$. The quotient-height inequality gives $h(C_0) \leq H$. For reduced $C_1 = u/v$, $v > 0$, the fraction $(u^2 + v^2)/v^2$ is reduced. Thus $h((1 + C_1^2)^2) = 2\log(u^2 + v^2) \geq 4h(C_1)$, and $h(C_1) \leq H/2$. Consequently $h(C) \leq 3H/2 + h(\rho) + \log 2$. Theorem 1.1 bounds each $h(x([n_i]P))$ by $h(C)$. The standard comparison $h(x([n]P)) = 2n^2\widehat{h}(P) + O_E(1)$, with $\widehat{h}(P) > 0$, proves the result; see [2, Chapter VIII, Section 9]. □

This extends [1, Corollary 2.4]. It does not permit unrestricted indices in a product. An aggregate theorem for a *sum* of abscissas is a different statement and is not contradicted by the example below.

6 An exact example and verification

Direct rational arithmetic gives

$$x(P) = \frac{57}{8} - \frac{7}{2}\rho, \qquad x(2P) = \frac{8226963211}{2285245536} - \frac{1063929023}{1713934152}\rho.$$

At the prime $\mathfrak{p} = (41, \rho - 24)$ their orders are $2, -2$; at the conjugate prime $(41, \rho - 17)$ both orders are zero. Moreover

$$\mathrm{N}_{K/\mathbb{Q}}\, x(P) = \frac{41^2}{2^6}.$$

The denominators of $x(P)$ are supported only above 2. Thus its only positive valuation is the order two just displayed. At (ρ) both abscissas have negative valuation, and both real embeddings are positive and greater than one. The defect formula proves the exact identity

$$h(x(P)) + h(x(2P)) - h(x(P)x(2P)) = \log 41.$$

This is a counterexample to deleting the common-valuation condition, not a counterexample to Sun's odd-index lemma.

The accompanying standard-library program uses exact fractions in K, checks the two abscissas by addition and duplication, and certifies the local orders by Hensel lifting with an explicit precision check. It also compares division-polynomial evaluation with repeated point addition. These finite computations verify the displayed example and identities; the proofs above establish the universal assertions.

References

[1] Zhi-Wei Sun, $\mathbb{Z}$ *is Diophantine over* $\mathbb{Q}$ *with 28 unknowns*, arXiv:2611.14605v1, 10 November 2026, https://arxiv.org/html/2611.14605v1, Lemma 2.2 and Corollary 2.4.

[2] Joseph H. Silverman, *The Arithmetic of Elliptic Curves*, second edition, Graduate Texts in Mathematics 106, Springer, 2009, Chapters IV, VII and VIII. DOI: 10.1007/978-0-387-09494-6.